

Causal Survival - Case Base

2026-05-31

Imports

EDA

```
library("Hmisc")  
library("tableone")  
library("xtable")  
library("ggplot2")  
library("survey")  
library("cobalt")
```

Estimation

```
library("survival")  
library("casebase")  
library("sandwich")  
library("foreach")  
library("doParallel")
```

EDA

Data Set

```
# set seed
s = 123

# load the dataset
getHdata(rhc)

# re-factor the data set

## main variables of interest

### time: last contact/death date - start date
time <- as.numeric(pmax(rhc$dthdte, rhc$lstctdte, na.rm = TRUE) - rhc$sadmdte)

### event: death
Delta <- ifelse(
  rhc$death == "Yes",
  1,
  0
)

### treatment: rhc performed or not
A <- ifelse(
  rhc$swangl == "RHC",
  1,
  0
)

## covariates

### demographics
Gender <- rhc$sex
Age <- as.numeric(rhc$age)

# contains 0 instead of NA
# changing 0 to NA completely messes up propensity scores
# and covariate balance after weighting
Weight <- as.numeric(rhc$wtkilo1)

Race <- droplevels(rhc$race)
Education <- as.numeric(rhc$edu)
Income <- rhc$income
InsuranceClass <- rhc$ninsclas

### comorbidities
hx.vars <- c(
  "chf hx",
  "cardio hx",
  "dement hx",
```

```

"psychhx",
"chrpulhx",
"renalhx",
"liverhx",
"gibledhx",
"malignhx",
"immunhx",
"transhx",
"amihx"
)

### convert labelled data to factors
rhc[hx.vars] <- lapply(rhc[hx.vars], function(x) {
  x <- as.numeric(x)

  factor(ifelse(x == 1, "Yes", "No"),
    levels = c("No", "Yes"))
})

CHFhx <- rhc$chfhx
CARDIOhx <- rhc$cardiohx
DEMENThx <- rhc$dementhx
PSYCHhx <- rhc$psychhx
CHRPULhx <- rhc$chrpulhx
RENALhx <- rhc$renalhx
LIVERhx <- rhc$liverhx
GIBLEDhx <- rhc$gibledhx
MALIGHx <- rhc$malignhx
IMMUNhx <- rhc$immunhx
TRANSHx <- rhc$transhx
AMIHx <- rhc$amihx

# severities
Cancer <- rhc$ca
Category <- droplevels(rhc$cat1)
Severity <- as.numeric(rhc$das2d3pc)

# data set design
data.set <- data.frame(
  time = time,
  Delta = Delta,
  A = A,

  Gender = Gender,
  Age = Age,
  Weight = Weight,
  Race = Race,
  Education = Education,
  Income = Income,
  InsuranceClass = InsuranceClass,

  CHFhx = CHFhx,
  CARDIOhx = CARDIOhx,

```

```
DEMENThx = DEMENThx,  
PSYCHhx = PSYCHhx,  
CHRPULhx = CHRPULhx,  
RENALhx = RENALhx,  
LIVERhx = LIVERhx,  
GIBLEDhx = GIBLEDhx,  
MALIGHx = MALIGHx,  
IMMUNhx = IMMUNhx,  
TRANShx = TRANShx,  
AMIhx = AMIhx,  
  
Cancer = Cancer,  
Category = Category,  
Severity = Severity  
)
```

Table One

```

covariates = c(
  "Gender",
  "Age",
  "Weight",
  "Race",
  "Education",
  "Income",
  "InsuranceClass",

  "CHFhx",
  "CARDIOhx",
  "DEMENThx",
  "PSYCHhx",
  "CHRPULhx",
  "RENALhx",
  "LIVERhx",
  "GIBLEDhx",
  "MALIGHx",
  "IMMUNhx",
  "TRANShx",
  "AMIhx",

  "Cancer",
  "Category",
  "Severity"
)

table.one.original <- CreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set,
  test = FALSE
)
print(table.one.original, smd = TRUE)

```

		Stratified by A		
		0	1	SMD
##	n	3551	2184	
##	Gender = Male (%)	1914 (53.9)	1278 (58.5)	0.093
##	Age (mean (SD))	61.76 (17.29)	60.75 (15.63)	0.061
##	Weight (mean (SD))	65.04 (29.50)	72.36 (27.73)	0.256
##	Race (%)			0.036
##	white	2753 (77.5)	1707 (78.2)	
##	black	585 (16.5)	335 (15.3)	
##	other	213 (6.0)	142 (6.5)	
##	Education (mean (SD))	11.57 (3.13)	11.86 (3.16)	0.091
##	Income (%)			0.142
##	Under \$11k	2081 (58.6)	1145 (52.4)	
##	\$11-\$25k	713 (20.1)	452 (20.7)	
##	\$25-\$50k	500 (14.1)	393 (18.0)	
##	> \$50k	257 (7.2)	194 (8.9)	

##	InsuranceClass (%)				0.194
##	No insurance	186 (5.2)	136 (6.2)		
##	Private	967 (27.2)	731 (33.5)		
##	Medicare	947 (26.7)	511 (23.4)		
##	Medicaid	454 (12.8)	193 (8.8)		
##	Private & Medicare	746 (21.0)	490 (22.4)		
##	Medicare & Medicaid	251 (7.1)	123 (5.6)		
##	CHFhx = Yes (%)	596 (16.8)	425 (19.5)	0.070	
##	CARDIOhx = Yes (%)	567 (16.0)	446 (20.4)	0.116	
##	DEMENThx = Yes (%)	413 (11.6)	151 (6.9)	0.163	
##	PSYCHhx = Yes (%)	286 (8.1)	100 (4.6)	0.143	
##	CHRPULhx = Yes (%)	774 (21.8)	315 (14.4)	0.192	
##	RENALhx = Yes (%)	149 (4.2)	106 (4.9)	0.032	
##	LIVERhx = Yes (%)	265 (7.5)	136 (6.2)	0.049	
##	GIBLEDhx = Yes (%)	131 (3.7)	54 (2.5)	0.070	
##	MALIGHx = Yes (%)	872 (24.6)	444 (20.3)	0.101	
##	IMMUNhx = Yes (%)	907 (25.5)	636 (29.1)	0.080	
##	TRANShx = Yes (%)	335 (9.4)	327 (15.0)	0.170	
##	AMIHx = Yes (%)	105 (3.0)	95 (4.3)	0.074	
##	Cancer (%)			0.107	
##	No	2652 (74.7)	1727 (79.1)		
##	Yes	638 (18.0)	334 (15.3)		
##	Metastatic	261 (7.4)	123 (5.6)		
##	Category (%)			0.583	
##	ARF	1581 (44.5)	909 (41.6)		
##	COPD	399 (11.2)	58 (2.7)		
##	CHF	247 (7.0)	209 (9.6)		
##	Cirrhosis	175 (4.9)	49 (2.2)		
##	Coma	341 (9.6)	95 (4.3)		
##	Colon Cancer	6 (0.2)	1 (0.0)		
##	Lung Cancer	34 (1.0)	5 (0.2)		
##	MOSF w/Malignancy	241 (6.8)	158 (7.2)		
##	MOSF w/Sepsis	527 (14.8)	700 (32.1)		
##	Severity (mean (SD))	20.37 (5.48)	20.70 (5.03)	0.063	

Propensity Scores

```
# set seed for output reproducibility
set.seed(s)

# generate propensity scores
propensity.score <- fitted(
  glm(
    A ~ Gender +
      Age +
      Weight +
      Race +
      Education +
      Income +
      InsuranceClass +
      CHFhx +
      CARDIOhx +
      DEMENThx +
      PSYCHhx +
      CHRPULhx +
      RENALhx +
      LIVERhx +
      GIBLEDhx +
      MALIGHx +
      IMMUNhx +
      TRANShx +
      AMIhx +
      Cancer +
      Category +
      Severity,
    data = data.set,
    family = binomial(link = "logit")
  )
)

# separate propensity scores for exposed & unexposed groups
propensity.score.0 <- propensity.score[A == 0]
propensity.score.1 <- propensity.score[A == 1]

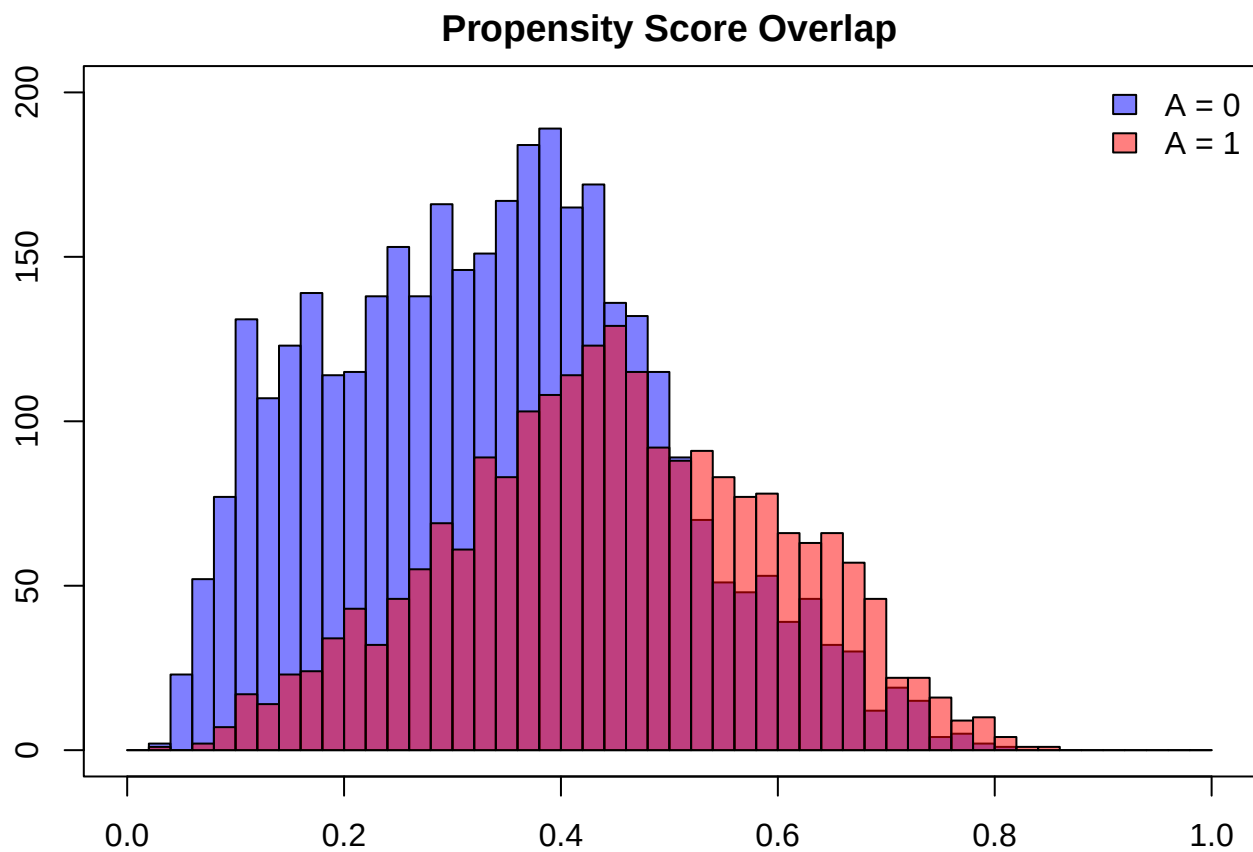
# visualize the propensity scores overlap
par(
  mar=c(2,2,2,0)
)

hist(
  propensity.score.0,
  col=rgb(0,0,1,0.5),
  breaks=seq(0,1,by=0.02),
  ylim=c(0,200),
  main="Propensity Score Overlap",
  xlab="Propensity Scores"
)
```

```
hist(
  propensity.score.1,
  col=rgb(1,0,0,0.5),
  breaks=seq(0,1,by=0.02),
  add=T
)

box()

legend(
  "topright",
  legend=c('A = 0', 'A = 1'),
  fill=c(rgb(0,0,1,0.5),rgb(1,0,0,0.5)),
  border="black",
  bty="n"
)
```



Stabilized IPW

```
p.A1 <- mean(A)

# calculate unstabilized ipws
ipw.unstabilized <- ifelse(
  A == 1,
  1 / propensity.score,
  1 / (1 - propensity.score)
)

# calculate stabilized ipws
ipw.stabilized <- ifelse(
  A == 1,
  p.A1 / propensity.score,
  (1 - p.A1) / (1 - propensity.score)
)

# visualize the ipws
par(
  mar=c(2,2,2,0)
)

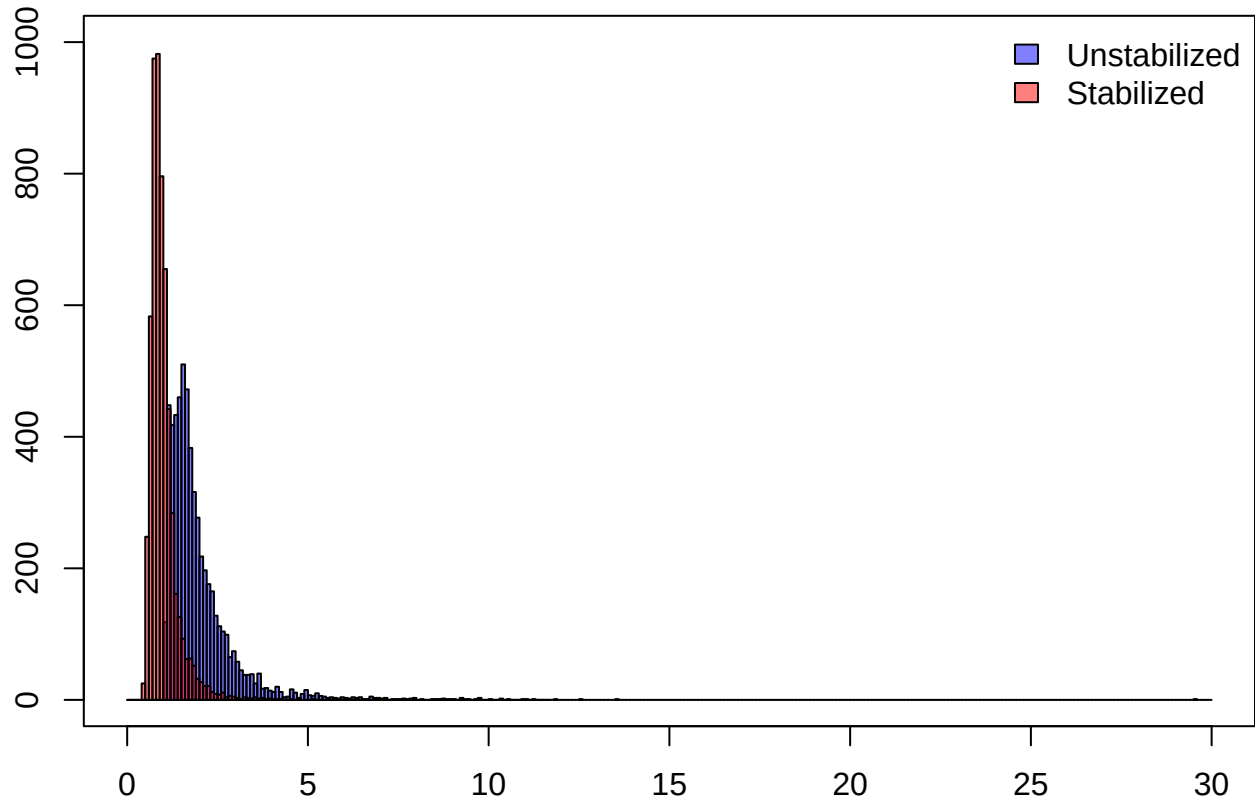
hist(
  ipw.unstabilized,
  col=rgb(0,0,1,0.5),
  breaks=seq(0, max(ipw.unstabilized) + 0.5,by=0.1),
  ylim=c(0,1000),
  main="Unstabilized Vs. Stabilized IPWs",
  xlab="IPWs"
)

hist(
  ipw.stabilized,
  col=rgb(1,0,0,0.5),
  breaks=seq(0, max(ipw.stabilized) + 0.5,by=0.1),
  add=T
)

box()

legend(
  "topright",
  legend=c('Unstabilized', 'Stabilized'),
  fill=c(rgb(0,0,1,0.5),rgb(1,0,0,0.5)),
  border="black", bty="n"
)
```

Unstablized Vs. Stabilized IPWs

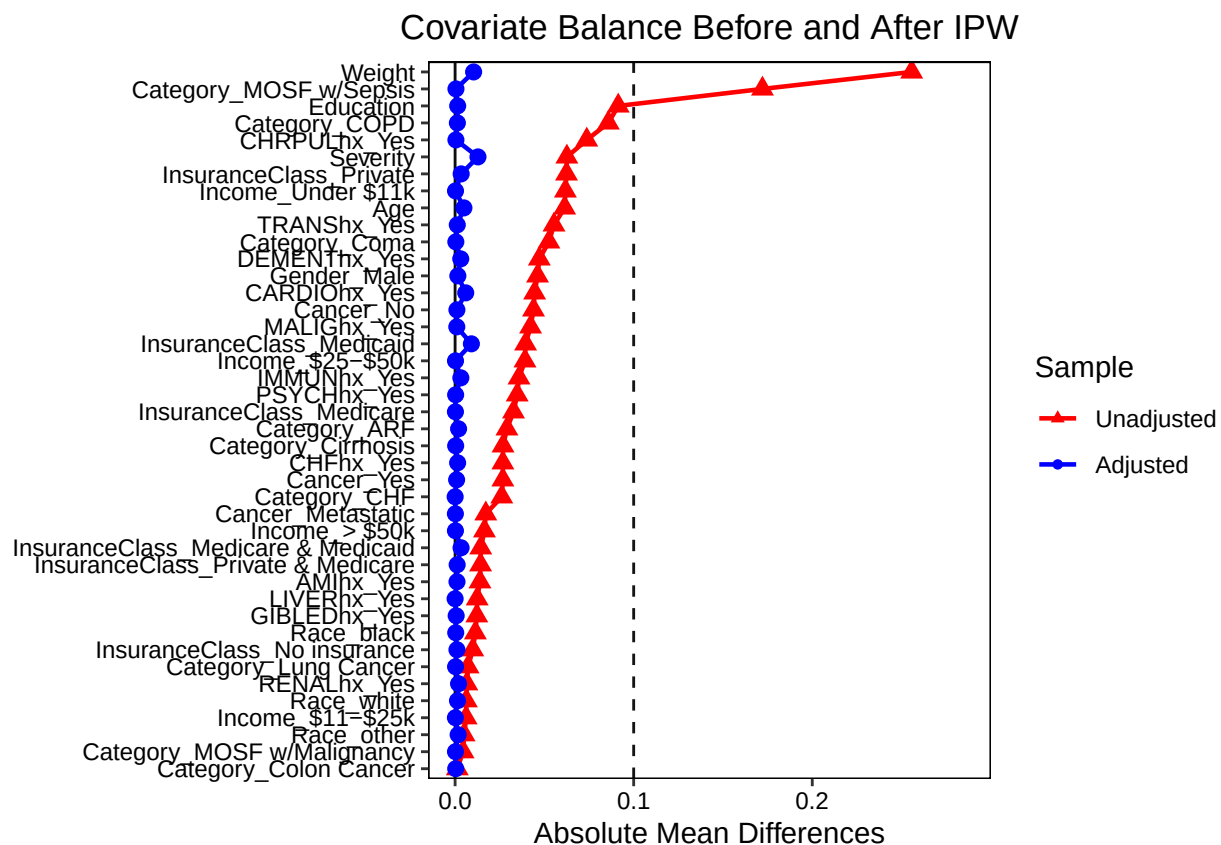


```
# ipw.stabilized is better so we augment that  
data.set$ipw <- ipw.stabilized
```

Covariate Balance

```
# calculate the covariate balance before & after IPW
balance <- bal.tab(
  A ~ Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  weights = ipw.stabilized,
  method = "weighting",
  estimand = "Average Treatment Effect",
  un = TRUE
)

# visualize the covariate balance before & after IPW
love.plot(
  balance,
  stats = "mean.diffs",
  threshold = 0.1,
  abs = TRUE,
  line = TRUE,
  var.order = "unadjusted",
  colors = c("red", "blue"),
  shapes = c(17, 16),
  limits = c(m = c(0, 0.3)),
  title = "Covariate Balance Before and After IPW",
  labels = TRUE
)
```



Note that covariate balance improved substantially after the inverse probability weighting (IPW) procedure was applied to create the pseudo-population.

Table Two

```
data.set.ipw <- svydesign(
  ids = ~ 0,
  data = data.set,
  weights = ipw.stabilized
)

table.one.ipw <- svyCreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set.ipw,
  test = FALSE
)

print(table.one.ipw, smd = TRUE)
```

	Stratified by A		
	0	1	SMD
n	3554.6	2178.4	
Gender = Male (%)	1968.2 (55.4)	1209.5 (55.5)	0.003
Age (mean (SD))	61.33 (17.24)	61.25 (15.74)	0.005
Weight (mean (SD))	67.93 (29.11)	68.23 (28.44)	0.010
Race (%)			0.007
white	2759.7 (77.6)	1694.1 (77.8)	
black	576.6 (16.2)	354.1 (16.3)	
other	218.3 (6.1)	130.1 (6.0)	
Education (mean (SD))	11.69 (3.14)	11.69 (3.08)	0.001
Income (%)			0.001
Under \$11k	1997.5 (56.2)	1224.6 (56.2)	
\$11-\$25k	723.4 (20.4)	443.7 (20.4)	
\$25-\$50k	555.6 (15.6)	340.0 (15.6)	
> \$50k	278.1 (7.8)	170.0 (7.8)	
InsuranceClass (%)			0.031
No insurance	198.0 (5.6)	119.1 (5.5)	
Private	1047.8 (29.5)	634.7 (29.1)	
Medicare	899.1 (25.3)	550.6 (25.3)	
Medicaid	404.3 (11.4)	267.6 (12.3)	
Private & Medicare	775.2 (21.8)	472.6 (21.7)	
Medicare & Medicaid	230.1 (6.5)	133.7 (6.1)	
CHFhx = Yes (%)	634.9 (17.9)	392.2 (18.0)	0.004
CARDIOhx = Yes (%)	636.8 (17.9)	403.2 (18.5)	0.015
DEMENThx = Yes (%)	347.5 (9.8)	206.1 (9.5)	0.011
PSYCHhx = Yes (%)	237.3 (6.7)	146.1 (6.7)	0.001
CHRPULhx = Yes (%)	669.0 (18.8)	411.0 (18.9)	0.001
RENALhx = Yes (%)	161.7 (4.5)	103.3 (4.7)	0.009
LIVERhx = Yes (%)	250.2 (7.0)	153.3 (7.0)	<0.001
GIBLEDhx = Yes (%)	114.7 (3.2)	69.0 (3.2)	0.003
MALIGHx = Yes (%)	817.4 (23.0)	503.0 (23.1)	0.002
IMMUNhx = Yes (%)	965.2 (27.2)	598.5 (27.5)	0.007
TRANShx = Yes (%)	412.5 (11.6)	255.5 (11.7)	0.004
AMIhx = Yes (%)	122.4 (3.4)	77.3 (3.5)	0.006
Cancer (%)			0.002

##	No	2712.1 (76.3)	1659.9 (76.2)	
##	Yes	603.9 (17.0)	371.9 (17.1)	
##	Metastatic	238.5 (6.7)	146.5 (6.7)	
##	Category (%)			0.011
##	ARF	1541.0 (43.4)	948.6 (43.5)	
##	COPD	282.7 (8.0)	170.5 (7.8)	
##	CHF	282.7 (8.0)	173.3 (8.0)	
##	Cirrhosis	138.6 (3.9)	85.6 (3.9)	
##	Coma	270.2 (7.6)	164.7 (7.6)	
##	Colon Cancer	4.4 (0.1)	3.4 (0.2)	
##	Lung Cancer	24.1 (0.7)	14.2 (0.7)	
##	MOSF w/Malignancy	246.5 (6.9)	150.6 (6.9)	
##	MOSF w/Sepsis	764.4 (21.5)	467.4 (21.5)	
##	Severity (mean (SD))	20.48 (5.53)	20.55 (4.95)	0.013

Estimation

Assumptions

Non-Informative Censoring

##FSH: Modified Definition

Model 1: Unadjusted Cox

```
# set seed for output reproducibility
set.seed(s)
```

```
# fit model
model.1 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  robust = TRUE
)
```

```
# generate summary
summary(model.1)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, robust = TRUE)
##
##      n= 5735, number of events= 3722
##
##      coef exp(coef) se(coef) robust se      z Pr(>|z|)
## A 0.15198  1.16414  0.03351  0.03420 4.444 8.85e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## A      1.164      0.859      1.089      1.245
##
## Concordance= 0.527 (se = 0.005 )
## Likelihood ratio test= 20.35 on 1 df,  p=6e-06
## Wald test               = 19.75 on 1 df,  p=9e-06
## Score (logrank) test = 20.62 on 1 df,  p=6e-06,  Robust = 19.74  p=9e-06
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Case Base Setup

```
# set a high ratio for all casebase models
fsh.ratio <- 100
```

Model 2: Unadjusted Case Base

```
# set seed for output reproducibility
set.seed(s)

# fit model
model.2 <- fitSmoothHazard(
  Delta ~ A,
  data = data.set,
  ratio = fsh.ratio
)

# generate summary
summary(model.2)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 372200
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ A, data = data.set, ratio = fsh.ratio)
##
## Coefficients:
##           Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -5.73103    0.02125 -269.750  < 2e-16 ***
## A           0.18664    0.03364   5.548 2.89e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 41762  on 375921  degrees of freedom
## Residual deviance: 41732  on 375920  degrees of freedom
## AIC: 41736
##
## Number of Fisher Scoring iterations: 7
```

```
# HR
beta.2 <- coef(model.2)["A"]
se.2 <- sqrt(diag(vcovHC(model.2))["A"])
cat("HR-hat:", exp(beta.2), "\n") #glm
```

```
## HR-hat: 1.205197
```

```
cat("SE of HR-hat:", se.2, "\n")
```

```
## SE of HR-hat: 0.03364396
```

```
cat("95% CI: (", exp(beta.2 + c(-1,1)*1.96*se.2), ")\n")
```

```
## 95% CI: ( 1.128288 1.28735 )
```

Model 3: Covariate Adjusted Cox

```
# set seed for output reproducibility
set.seed(s)
```

```
# fit model
```

```
model.3 <- coxph(
  Surv(time, Delta) ~ A +
    Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  robust = TRUE
)
```

```
# generate summary
summary(model.3)
```

```
## Call:
```

```
## coxph(formula = Surv(time, Delta) ~ A + Gender + Age + Weight +
##       Race + Education + Income + InsuranceClass + CHFhx + CARDIOhx +
##       DEMENThx + PSYCHhx + CHRPULhx + RENALhx + LIVERhx + GIBLEDhx +
##       MALIGHx + IMMUNhx + TRANShx + AMIhx + Cancer + Category +
##       Severity, data = data.set, robust = TRUE)
```

```
##
```

```
## n= 5735, number of events= 3722
```

```
##
```

	coef	exp(coef)	se(coef)	robust se
## A	0.2533449	1.2883275	0.0357613	0.0376554
## GenderMale	0.1233932	1.1313292	0.0345960	0.0361201
## Age	0.0106143	1.0106709	0.0014473	0.0015407
## Weight	-0.0020044	0.9979976	0.0006072	0.0006244
## Raceblack	0.0205835	1.0207968	0.0480552	0.0491244
## Raceother	0.1232340	1.1311490	0.0707914	0.0682031

## Education	0.0054062	1.0054208	0.0060673	0.0062343
## Income\$11-\$25k	-0.1325086	0.8758954	0.0458674	0.0467457
## Income\$25-\$50k	-0.2138434	0.8074749	0.0562912	0.0575050
## Income> \$50k	-0.1785389	0.8364916	0.0744784	0.0769455
## InsuranceClassPrivate	-0.0360729	0.9645700	0.0834823	0.0830787
## InsuranceClassMedicare	-0.0569048	0.9446840	0.0854888	0.0853701
## InsuranceClassMedicaid	-0.1679833	0.8453679	0.0931216	0.0941160
## InsuranceClassPrivate & Medicare	-0.0871831	0.9165093	0.0873752	0.0877396
## InsuranceClassMedicare & Medicaid	-0.1063721	0.8990900	0.0996189	0.0989352
## CHFhxYes	0.0915364	1.0958567	0.0505137	0.0505770
## CARDIOhxYes	-0.0508656	0.9504064	0.0480933	0.0533188
## DEMENThxYes	0.1334706	1.1427877	0.0544424	0.0595816
## PSYCHhxYes	-0.1652208	0.8477065	0.0704540	0.0691374
## CHRPUHxYes	0.0935860	1.0981051	0.0504644	0.0489547
## RENALhxYes	0.1684734	1.1834968	0.0790452	0.0854669
## LIVERhxYes	0.6040056	1.8294320	0.0941283	0.1033602
## GIBLEDhxYes	-0.0280961	0.9722950	0.1159686	0.1252369
## MALIGHxYes	-0.0408440	0.9599789	0.1780595	0.1854789
## IMMUNhxYes	0.0621781	1.0641519	0.0382973	0.0399256
## TRANShxYes	0.0220129	1.0222570	0.0559133	0.0567055
## AMIhxYes	-0.0135365	0.9865547	0.0917747	0.0843918
## CancerYes	0.3383995	1.4027007	0.1801795	0.1875629
## CancerMetastatic	0.5557491	1.7432464	0.2045181	0.2086943
## CategoryCOPD	-0.2602465	0.7708615	0.0775658	0.0696279
## CategoryCHF	-0.3735585	0.6882807	0.0787545	0.0751274
## CategoryCirrhosis	0.0548676	1.0564007	0.1195645	0.1318199
## CategoryComa	1.0743080	2.9279660	0.0617195	0.0740730
## CategoryColon Cancer	-0.3422784	0.7101504	0.5064276	0.3328934
## CategoryLung Cancer	0.2756386	1.3173716	0.1831278	0.1423742
## CategoryMOSF w/Malignancy	0.6104038	1.8411746	0.0748845	0.0775313
## CategoryMOSF w/Sepsis	0.1153416	1.1222568	0.0451955	0.0480631
## Severity	-0.0297207	0.9707166	0.0034800	0.0033148
##	z Pr(> z)			
## A	6.728	1.72e-11	***	
## GenderMale	3.416	0.000635	***	
## Age	6.889	5.60e-12	***	
## Weight	-3.210	0.001327	**	
## Raceblack	0.419	0.675211		
## Raceother	1.807	0.070783	.	
## Education	0.867	0.385851		
## Income\$11-\$25k	-2.835	0.004587	**	
## Income\$25-\$50k	-3.719	0.000200	***	
## Income> \$50k	-2.320	0.020323	*	
## InsuranceClassPrivate	-0.434	0.664143		
## InsuranceClassMedicare	-0.667	0.505049		
## InsuranceClassMedicaid	-1.785	0.074285	.	
## InsuranceClassPrivate & Medicare	-0.994	0.320390		
## InsuranceClassMedicare & Medicaid	-1.075	0.282299		
## CHFhxYes	1.810	0.070320	.	
## CARDIOhxYes	-0.954	0.340088		
## DEMENThxYes	2.240	0.025082	*	
## PSYCHhxYes	-2.390	0.016860	*	
## CHRPUHxYes	1.912	0.055916	.	
## RENALhxYes	1.971	0.048699	*	

```

## LIVERhxYes          5.844 5.11e-09 ***
## GIBLEDhxYes        -0.224 0.822490
## MALIGHxYes         -0.220 0.825709
## IMMUNhxYes         1.557 0.119388
## TRANShxYes         0.388 0.697870
## AMIhxYes           -0.160 0.872566
## CancerYes          1.804 0.071201 .
## CancerMetastatic   2.663 0.007745 **
## CategoryCOPD       -3.738 0.000186 ***
## CategoryCHF        -4.972 6.62e-07 ***
## CategoryCirrhosis  0.416 0.677241
## CategoryComa       14.503 < 2e-16 ***
## CategoryColon Cancer -1.028 0.303859
## CategoryLung Cancer 1.936 0.052866 .
## CategoryMOSF w/Malignancy 7.873 3.46e-15 ***
## CategoryMOSF w/Sepsis 2.400 0.016404 *
## Severity           -8.966 < 2e-16 ***

```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

	exp(coef)	exp(-coef)	lower .95	upper .95
## A	1.2883	0.7762	1.1967	1.3870
## GenderMale	1.1313	0.8839	1.0540	1.2143
## Age	1.0107	0.9894	1.0076	1.0137
## Weight	0.9980	1.0020	0.9968	0.9992
## Raceblack	1.0208	0.9796	0.9271	1.1240
## Raceother	1.1311	0.8841	0.9896	1.2929
## Education	1.0054	0.9946	0.9932	1.0178
## Income\$11-\$25k	0.8759	1.1417	0.7992	0.9599
## Income\$25-\$50k	0.8075	1.2384	0.7214	0.9038
## Income> \$50k	0.8365	1.1955	0.7194	0.9727
## InsuranceClassPrivate	0.9646	1.0367	0.8196	1.1351
## InsuranceClassMedicare	0.9447	1.0586	0.7991	1.1167
## InsuranceClassMedicaid	0.8454	1.1829	0.7030	1.0166
## InsuranceClassPrivate & Medicare	0.9165	1.0911	0.7717	1.0885
## InsuranceClassMedicare & Medicaid	0.8991	1.1122	0.7406	1.0915
## CHFhxYes	1.0959	0.9125	0.9924	1.2101
## CARDIOhxYes	0.9504	1.0522	0.8561	1.0551
## DEMENThxYes	1.1428	0.8751	1.0168	1.2843
## PSYCHhxYes	0.8477	1.1797	0.7403	0.9707
## CHRPUHxYes	1.0981	0.9107	0.9976	1.2087
## RENALhxYes	1.1835	0.8450	1.0010	1.3993
## LIVERhxYes	1.8294	0.5466	1.4939	2.2403
## GIBLEDhxYes	0.9723	1.0285	0.7607	1.2428
## MALIGHxYes	0.9600	1.0417	0.6674	1.3808
## IMMUNhxYes	1.0642	0.9397	0.9841	1.1508
## TRANShxYes	1.0223	0.9782	0.9147	1.1424
## AMIhxYes	0.9866	1.0136	0.8362	1.1640
## CancerYes	1.4027	0.7129	0.9712	2.0259
## CancerMetastatic	1.7432	0.5736	1.1580	2.6242
## CategoryCOPD	0.7709	1.2972	0.6725	0.8836
## CategoryCHF	0.6883	1.4529	0.5940	0.7975
## CategoryCirrhosis	1.0564	0.9466	0.8159	1.3678
## CategoryComa	2.9280	0.3415	2.5323	3.3855

```

## CategoryColon Cancer          0.7102      1.4082      0.3698      1.3637
## CategoryLung Cancer           1.3174      0.7591      0.9966      1.7414
## CategoryMOSF w/Malignancy     1.8412      0.5431      1.5816      2.1433
## CategoryMOSF w/Sepsis         1.1223      0.8911      1.0214      1.2331
## Severity                      0.9707      1.0302      0.9644      0.9770
##
## Concordance= 0.663 (se = 0.005 )
## Likelihood ratio test= 893.8 on 38 df,  p=<2e-16
## Wald test                  = 880.2 on 38 df,  p=<2e-16
## Score (logrank) test = 982.3 on 38 df,  p=<2e-16,  Robust = 862.6 p=<2e-16
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).

```

Model 4: Covariate Adjusted Case Base

```
# set seed for output reproducibility
set.seed(s)
```

```
# fit model
```

```
model.4 <- fitSmoothHazard(
```

```
  Delta ~ A +
```

```
  Gender +
```

```
  Age +
```

```
  Weight +
```

```
  Race +
```

```
  Education +
```

```
  Income +
```

```
  InsuranceClass +
```

```
  CHFhx +
```

```
  CARDIOhx +
```

```
  DEMENThx +
```

```
  PSYCHhx +
```

```
  CHRPULhx +
```

```
  RENALhx +
```

```
  LIVERhx +
```

```
  GIBLEDhx +
```

```
  MALIGHx +
```

```
  IMMUNhx +
```

```
  TRANShx +
```

```
  AMIhx +
```

```
  Cancer +
```

```
  Category +
```

```
  Severity,
```

```
  data = data.set,
```

```
  ratio = fsh.ratio
```

```
)
```

```
# generate summary
```

```
summary(model.4)
```

```
## Fitting smooth hazards with case-base sampling
```

```
##
```

```
## Sample size: 5735
```

```
## Number of events: 3722
```

```
## Number of base moments: 372200
```

```
## ----
```

```
##
```

```
## Call:
```

```
## fitSmoothHazard(formula = Delta ~ A + Gender + Age + Weight +
```

```
##      Race + Education + Income + InsuranceClass + CHFhx + CARDIOhx +
```

```
##      DEMENThx + PSYCHhx + CHRPULhx + RENALhx + LIVERhx + GIBLEDhx +
```

```
##      MALIGHx + IMMUNhx + TRANShx + AMIhx + Cancer + Category +
```

```
##      Severity, data = data.set, ratio = fsh.ratio)
```

```
##
```

```
## Coefficients:
```

```
##
```

```
Estimate Std. Error z value Pr(>|z|)
```

```

## (Intercept)          -5.8915603  0.1665386 -35.377 < 2e-16 ***
## A                    0.3155658  0.0361814  8.722 < 2e-16 ***
## GenderMale           0.1210934  0.0348561  3.474 0.000513 ***
## Age                  0.0116840  0.0014625  7.989 1.36e-15 ***
## Weight               -0.0023179  0.0006146 -3.772 0.000162 ***
## Raceblack            0.0479816  0.0484723  0.990 0.322235
## Raceother            0.2273727  0.0715687  3.177 0.001488 **
## Education            0.0021454  0.0062014  0.346 0.729377
## Income$11-$25k       -0.1357087  0.0465550 -2.915 0.003557 **
## Income$25-$50k       -0.2514429  0.0572075 -4.395 1.11e-05 ***
## Income> $50k         -0.1297272  0.0759153 -1.709 0.087480 .
## InsuranceClassPrivate -0.0575884  0.0844939 -0.682 0.495512
## InsuranceClassMedicare -0.0610693  0.0864336 -0.707 0.479849
## InsuranceClassMedicaid -0.2033099  0.0939583 -2.164 0.030477 *
## InsuranceClassPrivate & Medicare -0.0994626  0.0885691 -1.123 0.261440
## InsuranceClassMedicare & Medicaid -0.1649922  0.1004331 -1.643 0.100423
## CHFhxYes             0.0726263  0.0511995  1.418 0.156046
## CARDIOhxYes          -0.0876867  0.0483877 -1.812 0.069960 .
## DEMENThxYes          0.1467983  0.0550910  2.665 0.007707 **
## PSYCHhxYes           -0.2045046  0.0709745 -2.881 0.003959 **
## CHRPULhxYes          0.1400016  0.0511172  2.739 0.006166 **
## RENALhxYes           0.1256602  0.0798614  1.573 0.115608
## LIVERhxYes           0.7165146  0.0957713  7.482 7.35e-14 ***
## GIBLEDhxYes          -0.0788358  0.1169877 -0.674 0.500387
## MALIGHxYes           -0.0491066  0.1818799 -0.270 0.787164
## IMMUNhxYes           0.0820650  0.0390467  2.102 0.035578 *
## TRANShxYes           0.0820785  0.0564777  1.453 0.146143
## AMIhxYes             0.0008858  0.0927177  0.010 0.992378
## CancerYes            0.3913855  0.1840516  2.126 0.033462 *
## CancerMetastatic     0.6683611  0.2118287  3.155 0.001604 **
## CategoryCOPD          -0.3594887  0.0781131 -4.602 4.18e-06 ***
## CategoryCHF           -0.4772165  0.0797600 -5.983 2.19e-09 ***
## CategoryCirrhosis     0.0643399  0.1223124  0.526 0.598868
## CategoryComa          1.3337153  0.0628611 21.217 < 2e-16 ***
## CategoryColon Cancer -0.3819698  0.5096275 -0.750 0.453551
## CategoryLung Cancer   0.3486835  0.1868307  1.866 0.061998 .
## CategoryMOSF w/Malignancy 0.7954671  0.0782103 10.171 < 2e-16 ***
## CategoryMOSF w/Sepsis 0.1335503  0.0454710  2.937 0.003314 **
## Severity              -0.0350405  0.0035045 -9.999 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 41762 on 375921 degrees of freedom
## Residual deviance: 40494 on 375883 degrees of freedom
## AIC: 40572
##
## Number of Fisher Scoring iterations: 8

```

```

# HR
beta.4 <- coef(model.4)["A"]
se.4 <- sqrt(diag(vcovHC(model.4))["A"])
cat("HR-hat:", exp(beta.4), "\n") #glm

```

```
## HR-hat: 1.371035
```

```
cat("SE of HR-hat:", se.4, "\n")
```

```
## SE of HR-hat: 0.036884
```

```
cat("95% CI: (", exp(beta.4 + c(-1,1)*1.96*se.4), ") \n")
```

```
## 95% CI: ( 1.275417 1.473821 )
```

Model 5: Marginal Structural Cox

```
# set seed for output reproducibility
set.seed(s)
```

```
# fit model
model.5 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  weights = ipw.stabilized,
  robust = TRUE
)
```

```
# generate summary
summary(model.5)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, weights = ipw.stabilized,
##       robust = TRUE)
##
##      n= 5735, number of events= 3722
##
##      coef exp(coef) se(coef) robust se      z Pr(>|z|)
## A 0.20890   1.23232  0.03338   0.03764 5.55 2.86e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## A      1.232      0.8115      1.145      1.327
##
## Concordance= 0.535 (se = 0.005 )
## Likelihood ratio test= 38.61 on 1 df,  p=5e-10
## Wald test               = 30.8 on 1 df,  p=3e-08
## Score (logrank) test = 39.32 on 1 df,  p=4e-10, Robust = 30.69 p=3e-08
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Model 6: IPW Case Base PH

```
# set seed for output reproducibility
set.seed(s)
```

```
# fit model
model.6 <- FSH(
  Delta ~ A,
  data = data.set,
  ratio = fsh.ratio,
  w = ipw.stabilized
)
```

```
# generate summary
summary(model.6)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 372200
## ----
##
## Call:
## FSH(formula = Delta ~ A, data = data.set, w = ipw.stabilized,
##      ratio = fsh.ratio)
##
## Coefficients:
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -5.75524     0.02141 -268.818  < 2e-16 ***
## A           0.24268     0.03351   7.242 4.41e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 41800  on 375921  degrees of freedom
## Residual deviance: 41748  on 375920  degrees of freedom
## AIC: 45713
##
## Number of Fisher Scoring iterations: 7
```

```
# HR
beta.6 <- coef(model.6)["A"]
se.6 <- sqrt(diag(vcovHC(model.6))["A"])
cat("HR-hat:", exp(beta.6), "\n") #glm
```

```
## HR-hat: 1.274666
```

```
cat("SE of HR-hat:", se.6, "\n")
```

```
## SE of HR-hat: 0.03693444
```

```
cat("95% CI: (", exp(beta.6 + c(-1,1)*1.96*se.6), ")\n")
```

```
## 95% CI: ( 1.185652 1.370363 )
```

Model 7: Unadjusted Case Base (Variable HR)

Generate Model Summary

```
# set seed for output reproducibility
set.seed(s)

# fit model
model.7 <- FSH(
  Delta ~ pspline(time, df = 2) * A,
  data = data.set,
  ratio = fsh.ratio
)

# generate summary
summary(model.7)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 372200
## ----
##
## Call:
## FSH(formula = Delta ~ pspline(time, df = 2) * A, data = data.set,
##      ratio = fsh.ratio)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      23.9106     0.9581  24.955 <2e-16 ***
## pspline(time, df = 2)1 -34.7856     1.1936 -29.144 <2e-16 ***
## pspline(time, df = 2)2 -28.4246     0.8358 -34.007 <2e-16 ***
## pspline(time, df = 2)3 -31.3457     1.0915 -28.718 <2e-16 ***
## pspline(time, df = 2)4 -28.6617     0.9190 -31.188 <2e-16 ***
## pspline(time, df = 2)5 -30.4740     1.2280 -24.817 <2e-16 ***
## pspline(time, df = 2)6 -26.1270     2.0543 -12.718 <2e-16 ***
## pspline(time, df = 2)7 -27.4090    22.5995  -1.213  0.2252
## A                3.1786     1.5638   2.033  0.0421 *
## pspline(time, df = 2)1:A -3.5115     1.9554  -1.796  0.0725 .
## pspline(time, df = 2)2:A -3.0311     1.3562  -2.235  0.0254 *
## pspline(time, df = 2)3:A -3.4688     1.7963  -1.931  0.0535 .
## pspline(time, df = 2)4:A -3.7057     1.5048  -2.463  0.0138 *
## pspline(time, df = 2)5:A -2.5620     1.9696  -1.301  0.1933
## pspline(time, df = 2)6:A -5.8423     3.0026  -1.946  0.0517 .
## pspline(time, df = 2)7:A  3.2765    25.6504   0.128  0.8984
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 41762 on 375921 degrees of freedom
## Residual deviance: 38170 on 375906 degrees of freedom
```

```
## AIC: 38202
##
## Number of Fisher Scoring iterations: 8
```

Setup for Survival Curve

```
trt.types <- c(0, 1)
individual.profile.data <- data.frame(A = 1)
survival.curves.data <- data.frame(A = trt.types)

tau.max <- max(data.set$time, na.rm = TRUE)
t.max <- 150 # First 4 months

tau.grid <- seq(0, tau.max, by = 1)
t.grid <- seq(0, t.max, by = 1)
```

Plot variable HR

```
# visualize the hazard ratio over time
par(mfrow = c(2, 1))

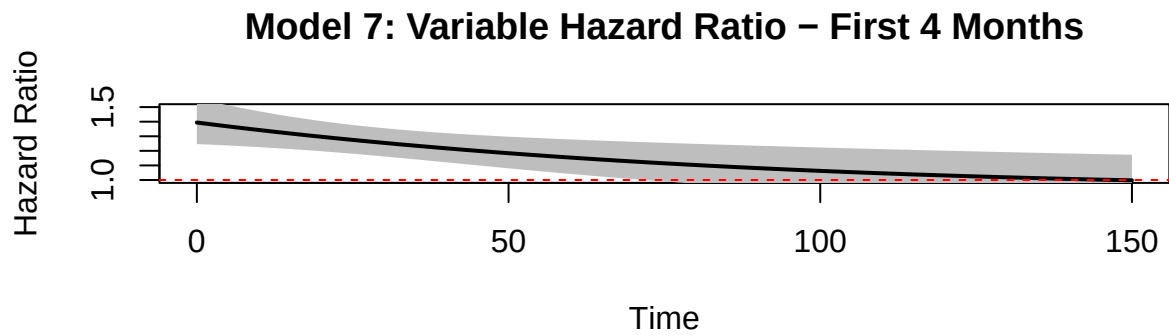
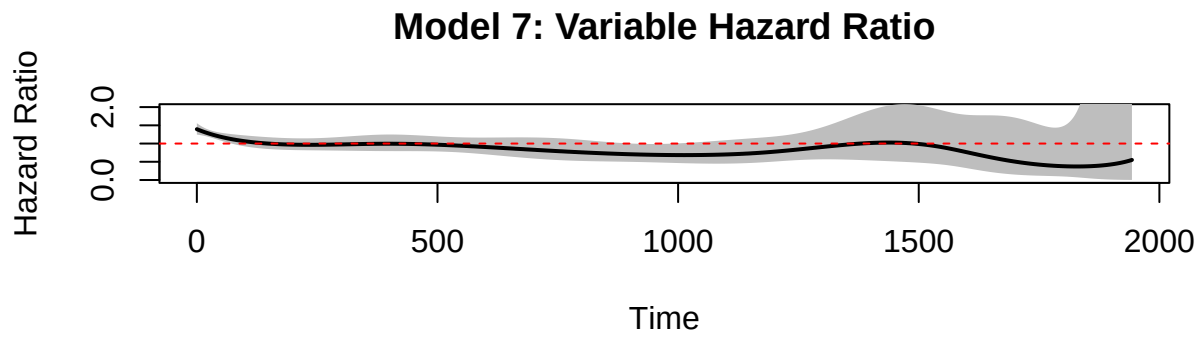
plot(
  model.7,
  type = "hr",
  newdata = expand.grid(
    time = tau.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(0,2),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 7: Variable Hazard Ratio"
)
abline(h = 1, col = "red", lty = 2)

plot(
  model.7,
  type = "hr",
  newdata = expand.grid(
    time = t.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(1,1.5),
```

```

xlab="Time",
ylab="Hazard Ratio",
main="Model 7: Variable Hazard Ratio - First 4 Months"
)
abline(h = 1, col = "red", lty = 2)

```



Model 8: IPW Case Base (Variable HR) (v1)

Generate Model Summary

```
# set seed for output reproducibility
set.seed(s)
```

```
# fit model
model.8 <- FSH(
  Delta ~ pspline(time, df = 2) * A,
  data = data.set,
  ratio = fsh.ratio,
  w = ipw.stabilized
)
```

```
# generate summary
summary(model.8)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 372200
## ----
##
## Call:
## FSH(formula = Delta ~ pspline(time, df = 2) * A, data = data.set,
##      w = ipw.stabilized, ratio = fsh.ratio)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      24.1612    0.9664  25.000  <2e-16 ***
## pspline(time, df = 2)1 -35.1315    1.2040 -29.178  <2e-16 ***
## pspline(time, df = 2)2 -28.6482    0.8430 -33.983  <2e-16 ***
## pspline(time, df = 2)3 -31.6778    1.1013 -28.764  <2e-16 ***
## pspline(time, df = 2)4 -28.8377    0.9263 -31.131  <2e-16 ***
## pspline(time, df = 2)5 -30.8929    1.2374 -24.966  <2e-16 ***
## pspline(time, df = 2)6 -26.0588    2.0324 -12.822  <2e-16 ***
## pspline(time, df = 2)7 -29.5572   22.5389  -1.311    0.1897
## A                2.4902    1.5557   1.601    0.1094
## pspline(time, df = 2)1:A -2.5903    1.9449  -1.332    0.1829
## pspline(time, df = 2)2:A -2.3605    1.3496  -1.749    0.0803 .
## pspline(time, df = 2)3:A -2.8199    1.7842  -1.580    0.1140
## pspline(time, df = 2)4:A -2.4443    1.4919  -1.638    0.1013
## pspline(time, df = 2)5:A -2.8973    1.9855  -1.459    0.1445
## pspline(time, df = 2)6:A -3.0804    2.9253  -1.053    0.2923
## pspline(time, df = 2)7:A -5.5768   25.7810  -0.216    0.8287
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 41800  on 375921  degrees of freedom
```

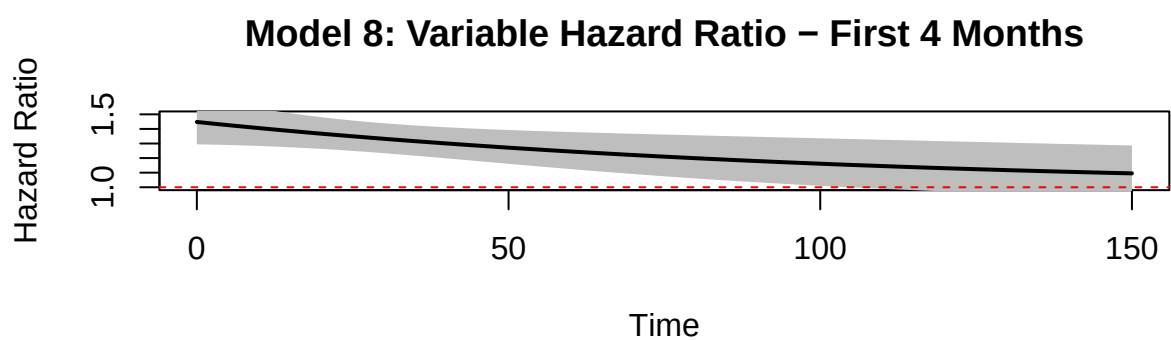
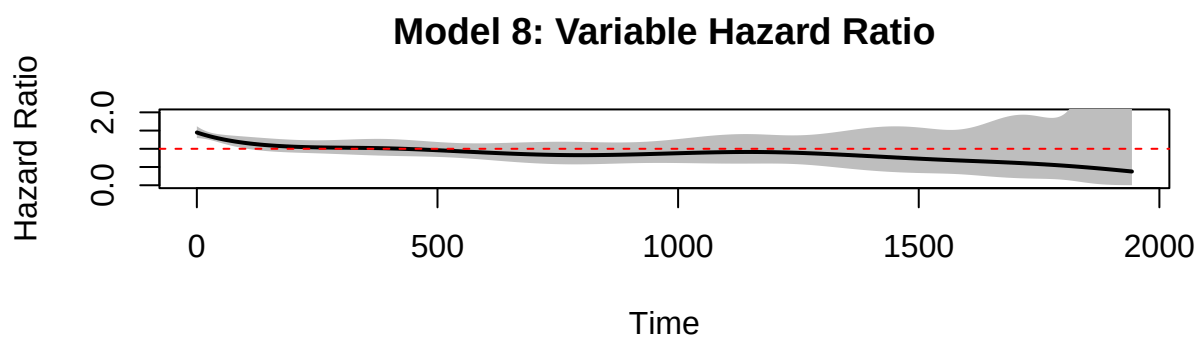
```
## Residual deviance: 38183  on 375906  degrees of freedom
## AIC: 41812
##
## Number of Fisher Scoring iterations: 8
```

Plot variable HR

```
# visualize the hazard ratio over time
par(mfrow = c(2, 1))

plot(
  model.8,
  type = "hr",
  newdata = expand.grid(
    time = tau.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(0,2),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 8: Variable Hazard Ratio"
)
abline(h = 1, col = "red", lty = 2)

plot(
  model.8,
  type = "hr",
  newdata = expand.grid(
    time = t.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(1,1.5),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 8: Variable Hazard Ratio - First 4 Months"
)
abline(h = 1, col = "red", lty = 2)
```



Survival Curves using MS-Cox and IPW Case Base Models

SetUp

```
# colors: blue for MS-Cox, red for case base (with matching transparent fills)
col.cox <- "blue"
col.cb  <- "red"

fill.cox <- rgb(0, 0, 1, 0.20)
fill.cb  <- rgb(1, 0, 0, 0.20)
```

Generate MS-Cox Survival Curve

```
sf.cox.ind <- survfit(
  model.5,
  newdata = individual.profile.data
)

S.cox <- approx(
  sf.cox.ind$time,
  sf.cox.ind$surv,
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

Lo.cox <- approx(
  sf.cox.ind$time,
  sf.cox.ind$lower,
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

Up.cox <- approx(
  sf.cox.ind$time,
  sf.cox.ind$upper,
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y
```

Generate Case Base Survival Curve

```
set.seed(s)
n.boot <- 200

boot.surv <- replicate(n.boot, {
```

```

idx <- sample(nrow(data.set), replace = TRUE)
boot.data <- data.set[idx, ]

boot.fit <- FSH(
  Delta ~ pspline(time, df = 2) * A,
  data = boot.data,
  ratio = 10,
  weights = boot.data$ipw
)

ar <- absoluteRisk(
  boot.fit,
  time = tau.grid,
  newdata = individual.profile.data,
  type = "survival"
)
ar[, 2]
})

S.cb <- absoluteRisk(
  model.8,
  time = tau.grid,
  newdata = individual.profile.data,
  type = "survival"
)[, 2]

Lo.cb <- apply(
  boot.surv,
  1,
  quantile,
  probs = 0.025,
  na.rm = TRUE
)

Up.cb <- apply(
  boot.surv,
  1,
  quantile,
  probs = 0.975,
  na.rm = TRUE
)

```

Plot Survival Curves

```

par(mfrow = c(2, 1))

plot(
  tau.grid,
  S.cox,
  type = "n",
  ylim = c(0, 1),

```

```

xlab = "Days",
ylab = "Survival Probability",
main = "Estimated Survival Probability After RHC"
)

# shaded bands first, lines on top
polygon(
  c(tau.grid, rev(tau.grid)),
  c(Lo.cox, rev(Up.cox)),
  col = fill.cox,
  border = NA
)

polygon(
  c(tau.grid, rev(tau.grid)),
  c(Lo.cb, rev(Up.cb)),
  col = fill.cb,
  border = NA
)

lines(tau.grid, S.cox, col = col.cox, lwd = 2, type = "s")
lines(tau.grid, S.cb, col = col.cb, lwd = 2)

legend(
  "topright",
  legend = c(
    "semi-parametric (MS-Cox)",
    "parametric (IPW Case Base)"
  ),
  col = c(col.cox, col.cb),
  lty = c(1, 1),
  bg = "gray90"
)

plot(
  tau.grid,
  S.cox,
  type = "n",
  ylim = c(0.6, 1),
  xlim = c(0, 30),
  xlab = "Days",
  ylab = "Survival Probability",
  main = "Estimated Survival Probability After RHC - First Month"
)

# shaded bands first, lines on top
polygon(
  c(tau.grid, rev(tau.grid)),
  c(Lo.cox, rev(Up.cox)),
  col = fill.cox,
  border = NA
)

```

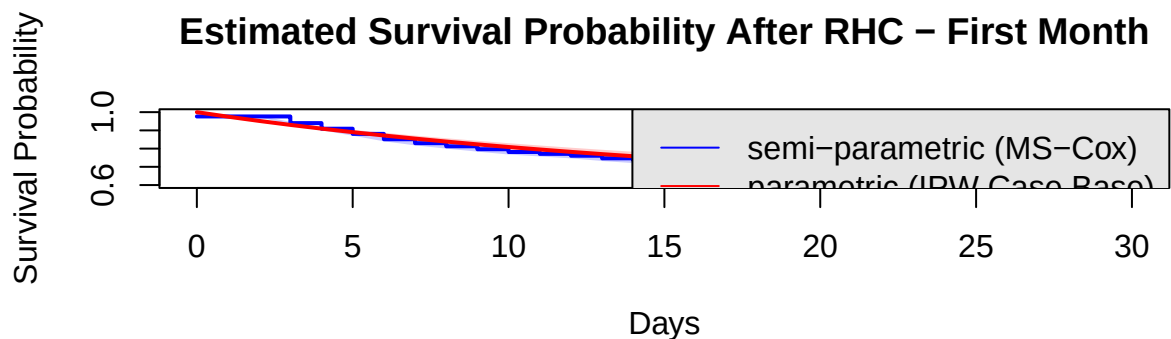
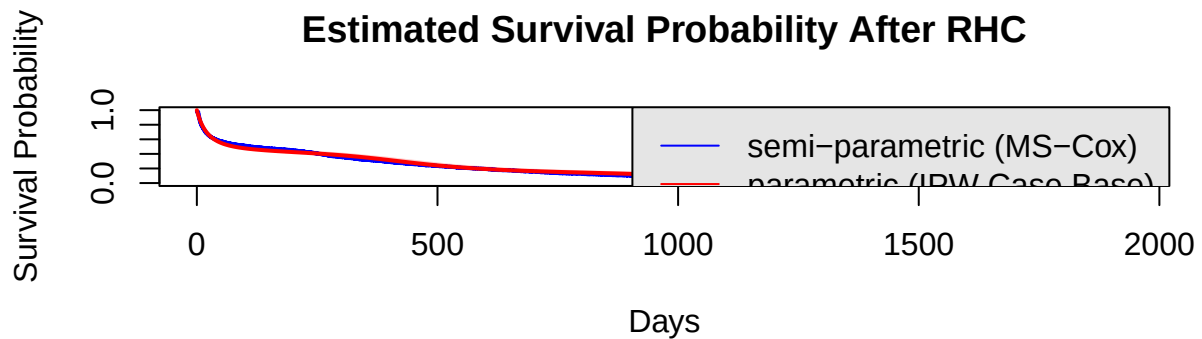
```

polygon(
  c(tau.grid, rev(tau.grid)),
  c(Lo.cb, rev(Up.cb)),
  col = fill.cb,
  border = NA
)

lines(tau.grid, S.cox, col = col.cox, lwd = 2, type = "s")
lines(tau.grid, S.cb, col = col.cb, lwd = 2)

legend(
  "topright",
  legend = c(
    "semi-parametric (MS-Cox)",
    "parametric (IPW Case Base)"
  ),
  col = c(col.cox, col.cb),
  lty = c(1, 1),
  bg = "gray90"
)

```



RMST Function

```
rmst.fn <- function(time, surv) {  
  dt <- diff(time)  
  mids <- (head(surv, -1) + tail(surv, -1)) / 2  
  c(0, cumsum(mids * dt))  
}
```

MS-Cox PH Counterfactual Survival Curves

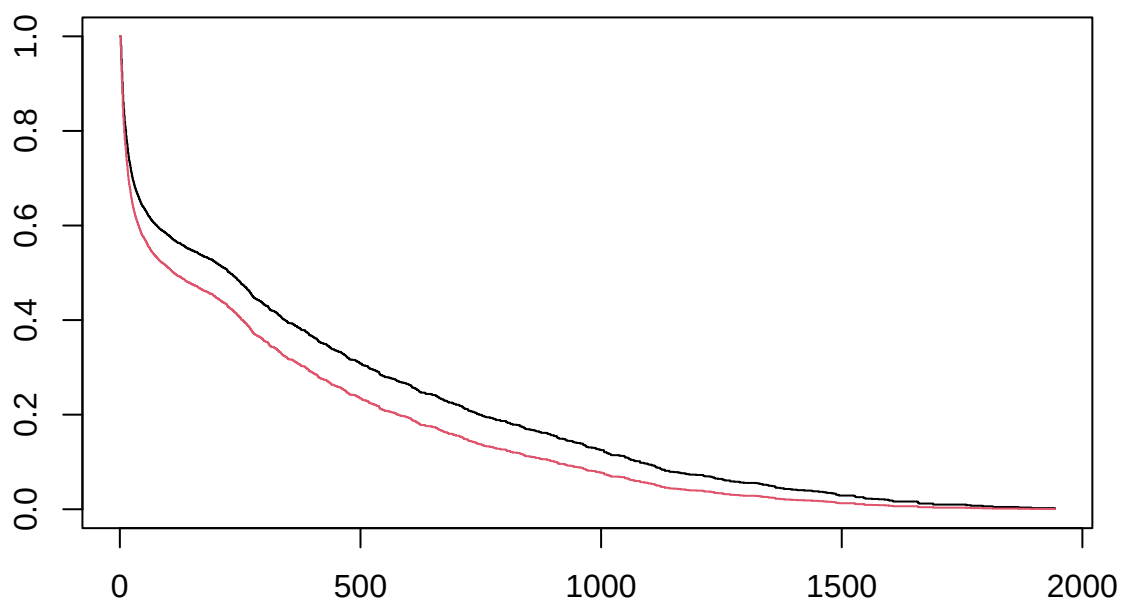
Extract MS-Cox Counterfactual Survival Curves

```
sf.cox <- survfit(  
  model.5,  
  newdata = survival.curves.data  
)
```

```
S0.cox <- approx(  
  sf.cox$time,  
  sf.cox$surv[, 1],  
  xout = tau.grid,  
  method = "constant",  
  rule = 2  
)$y
```

```
S1.cox <- approx(  
  sf.cox$time,  
  sf.cox$surv[, 2],  
  xout = tau.grid,  
  method = "constant",  
  rule = 2  
)$y
```

```
plot(sf.cox, col = c(1,2))
```



Plot MS-Cox Counterfactual Survival Curves

```
par(mfrow = c(2, 1))

plot(
  tau.grid,
  S0.cox,
  type = "s",
  lwd = 2,
  col = "red",
  ylim = c(0, 1),
  xlab = "Days",
  ylab = "Survival Probability",
  main = "MS-Cox: Counterfactual Survival Curves"
)

lines(
  tau.grid,
  S1.cox,
  type = "s",
  lwd = 2,
  col = "blue"
)
```

```

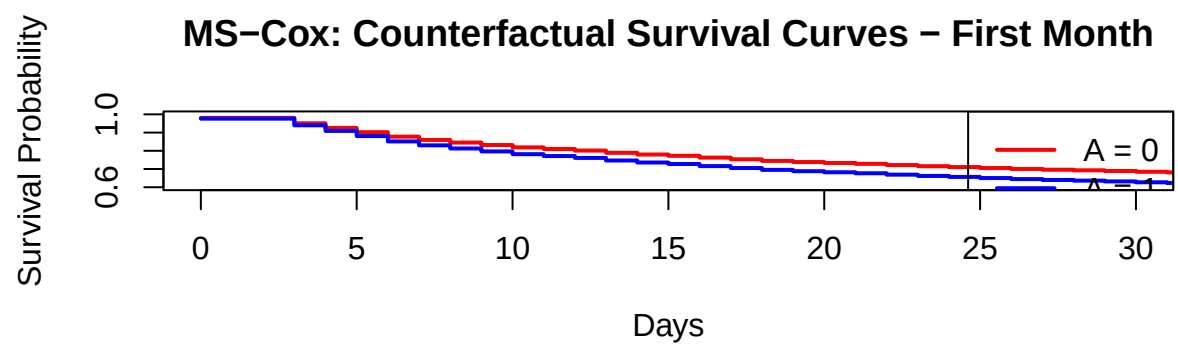
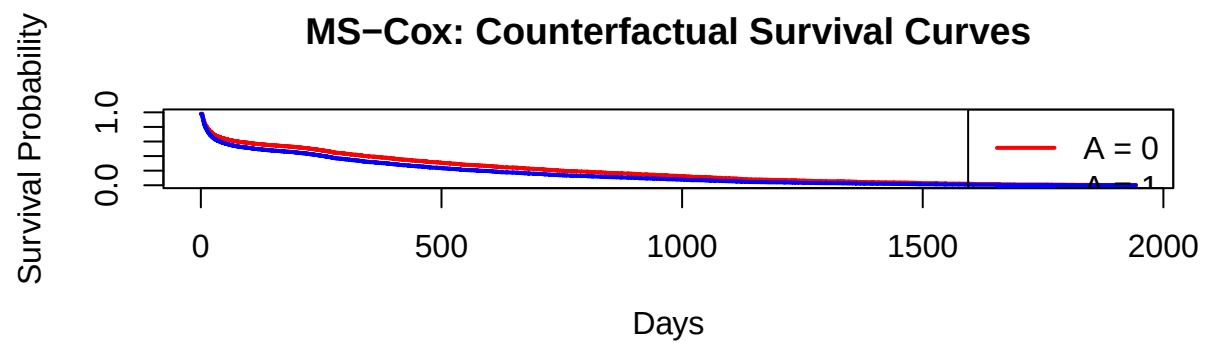
legend(
  "topright",
  legend = c("A = 0", "A = 1"),
  col = c("red", "blue"),
  lty = 1,
  lwd = 2
)

plot(
  tau.grid,
  S0.cox,
  type = "s",
  lwd = 2,
  col = "red",
  ylim = c(0.6, 1),
  xlim = c(0, 30),
  xlab = "Days",
  ylab = "Survival Probability",
  main = "MS-Cox: Counterfactual Survival Curves - First Month"
)

lines(
  tau.grid,
  S1.cox,
  type = "s",
  lwd = 2,
  col = "blue"
)

legend(
  "topright",
  legend = c("A = 0", "A = 1"),
  col = c("red", "blue"),
  lty = 1,
  lwd = 2
)

```



IPW Case Base Counterfactual Survival Curves

Extract CB Counterfactual Survival Curves

```
Sf.cb <- absoluteRisk(  
  object = model.8,  
  time = tau.grid,  
  newdata = survival.curves.data,  
  type = "survival"  
)  
  
S0.cb <- Sf.cb[, 2]  
S1.cb <- Sf.cb[, 3]
```

Plot CB Counterfactual Survival Curves

```
par(mfrow = c(2, 1))  
  
plot(  
  S0.cb,  
  type = "l",  
  lwd = 2,  
  col = "red",  
  ylim = c(0, 1),  
  xlab = "Days",  
  ylab = "Survival Probability",  
  main = "IPW Case Base: Counterfactual Survival Curves"  
)  
  
lines(  
  S1.cb,  
  lwd = 2,  
  col = "blue"  
)  
  
legend(  
  "topright",  
  legend = c("A = 0", "A = 1"),  
  col = c("red", "blue"),  
  lty = 1,  
  lwd = 2  
)  
  
plot(  
  S0.cb,  
  type = "l",  
  lwd = 2,  
  col = "red",  
  ylim = c(0.6, 1),  
  xlim = c(0, 30),  
  xlab = "Days",
```

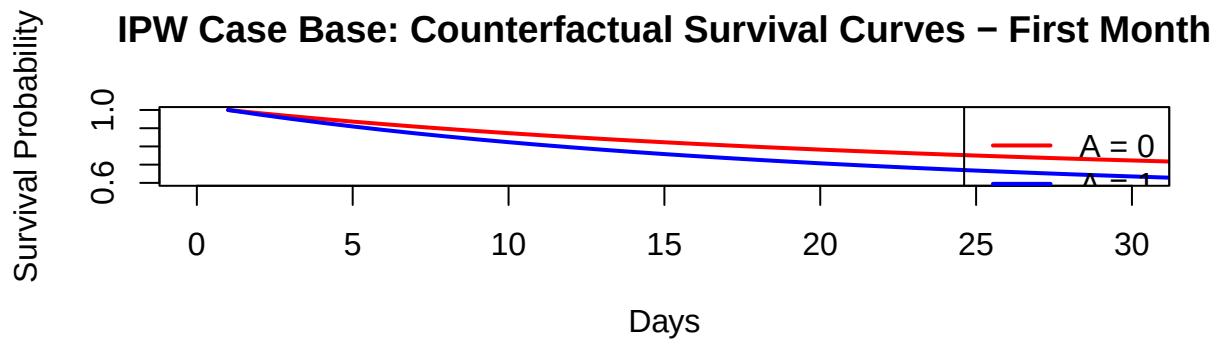
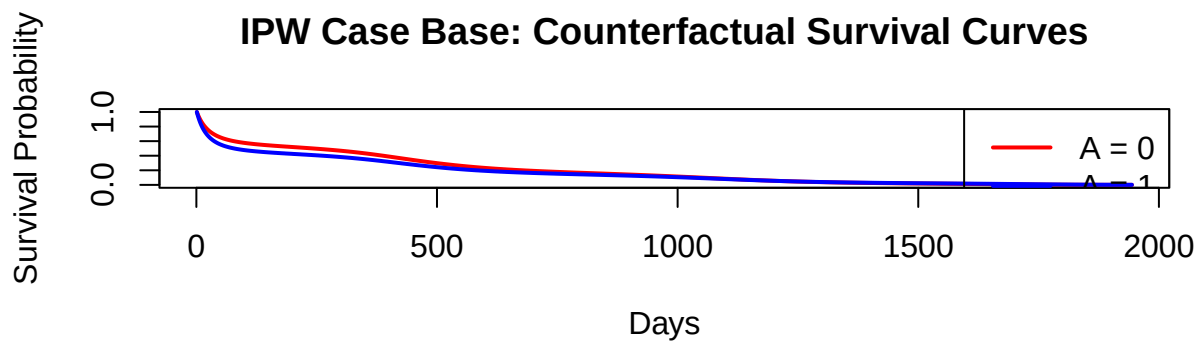
```

ylab = "Survival Probability",
main = "IPW Case Base: Counterfactual Survival Curves - First Month"
)

lines(
  S1.cb,
  lwd = 2,
  col = "blue"
)

legend(
  "topright",
  legend = c("A = 0", "A = 1"),
  col = c("red", "blue"),
  lty = 1,
  lwd = 2
)

```



RMST Comparison: MS-Cox vs. IPW Case Base

Calculate RMST for both models

```
RMST.cox <- rmst.fn(tau.grid, S1.cox) - rmst.fn(tau.grid, S0.cox)

RMST.cb <- rmst.fn(tau.grid, S1.cb) - rmst.fn(tau.grid, S0.cb)
```

Plot Both RMST

```
cols <- c("#8E063B", "#023FA5")

plot(
  RMST.cox,
  type = "l",
  lwd = 2,
  col = "red",
  xlab = "Tau (Days)",
  ylab = "RMST Difference",
  main = "Causal RMST up to Tau"
)

lines(
  RMST.cb,
  lwd = 2,
  col = "blue"
)

abline(h = 0, lty = 2)

legend(
  "topright",
  legend = c("MS-Cox", "IPW Case Base"),
  col = c("red", "blue"),
  lwd = 2,
  lty = 1
)
```

Causal RMST up to Tau

